



M003 Week 6 — English Worksheet (Advanced)

Topic: Recreate the Ender Dragon (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

A big blocky **ender dragon** stands in your world — black blocks, purple eyes. A big model is a set of **parts**: a long **body** (a fill box stretched along one axis), a **head** at the front, two **wings** that spread out along x, and a **tail** that steps back and down. Everything uses **(x, y, z)** — x **across**, y **up**, z **forward** (deep). Plan your corners before you build.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🧠

Read each code. Write your full prediction.

```
fill black from (0, 8, 0) to (0, 8, 10)
```

One block wide, stretched along z. How long is the body? Which way does it run?


```
fill black from (1, 8, 5) to (5, 8, 5)  
fill black from (-5, 8, 5) to (-1, 8, 5)
```

Two wings. Both stay at y = 8 and z = 5. How wide is each wing? Are they the same size on both sides?


```
place black block at (0, 7, -1)
place black block at (0, 6, -2)
place black block at (0, 5, -3)
```

**This is the tail behind the body. As z gets smaller (more negative), what happens to the height?
Across, up, or down-and-back?**

2 · Spot the Bug 🐛

Bug A — These should be **two matching wings**, spreading the same distance out on each side from the body at $x = 0$. Right now the left wing is much shorter than the right.

```
fill black from (1, 8, 5) to (5, 8, 5)
fill black from (-2, 8, 5) to (-1, 8, 5)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should be a **long body** stretched **deep along z**, from $(0, 8, 0)$ to $(0, 8, 10)$. Right now it lies flat across the ground along x instead.

```
fill black from (0, 8, 0) to (10, 8, 0)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The **tail** should step **back and down** behind the body: each block goes one deeper and one lower. Right now every tail block sits at the same height, so the tail floats straight out instead of drooping.

```
place black block at (0, 8, -1)
place black block at (0, 8, -2)
place black block at (0, 8, -3)
```

Hint: look at the **y** numbers. A tail that droops must lower y each step, not keep it the same.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Build the Full Dragon 🧱

Look at the dragon in your world. Walk around it. Plan your corners, then build it part by part on your home spot:

```
fill black from (0, 8, 0) to (0, 8, 10)
place black block at (0, 9, 11)
fill black from (1, 8, 5) to (5, 8, 5)
fill black from (-5, 8, 5) to (-1, 8, 5)
place black block at (0, 7, -1)
place black block at (0, 6, -2)
place black block at (0, 5, -3)
```

That is body, head, right wing, left wing, and a stepped tail. Add two **purple** eye blocks on the head. Keep the wings the same length on both sides.

Send a photo or video. Then finish these — 4 to 6 sentences.

I built the dragon as these parts: ...

My body runs along the ... axis, from (... , ... , ...) to (... , ... , ...).

To make the wings match, I ...

My tail steps back and down by changing the ... number and the ... number each time.

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Film yourself showing your dragon. Use these words: **x, y, z, body, head, wing, tail, home spot.**

1. Show your home spot. Why stand on it every run?
2. Point at the body. Say which axis it runs along and why.
3. Point at both wings. Show they are the same length, and say in your own words how you kept them matching.
4. Point at the tail. Say which numbers change to make it step back and down.

Plan what you will say:

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.